SaaS P2P Technical Integration Guide

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4. Introduction

This Technical Integration Guide provides detailed instructions for integrating the **SaaS-based Purchase-to-Pay (P2P) system** with enterprise backend systems, including ERP solutions, ETL pipelines, and API-based services. The guide is intended to assist system administrators, integration consultants, and developers in configuring, deploying, and validating the P2P application within cloud-based environments.

The primary purpose of this guide is to ensure a **standardized and reliable integration process** that enables seamless data exchange between the platform and connected systems. It covers the installation of necessary components, configuration of middleware adapters, API integration for invoicing, account management, and exchange rate synchronization, as well as deployment and validation of ETL processes. The guide also includes troubleshooting procedures for common integration issues and outlines best practices for maintaining data integrity and system security.

The scope of this document includes:

- Integration of the SaaS P2P system with middleware and ERP adapters.
- Configuration of API endpoints and service deployment on OSGi-based platforms.
- ETL procedures for data extraction, transformation, and loading into target systems.
- Validation of transactional data, notifications, and user requests.

The guide excludes system-level administration of the underlying ERP platform, network infrastructure setup, and custom development outside the standard integration framework.

At a high level, the integration system consists of the **cloud platform**, connected ERP systems, middleware adapters, and ETL pipelines. Services are deployed and managed in a centralized OSGi environment, while data exchange relies on secure API calls using JSON messages, ensuring accurate and timely synchronization of procurement, invoicing, and financial records.

5. System Overview

The **SaaS Purchase-to-Pay (P2P) system** is a cloud-based application designed to automate procurement, invoicing, and supplier management. It integrates seamlessly with enterprise backend systems, including ERP solutions, ETL pipelines, and reporting modules, ensuring accurate and timely financial operations.

System Architecture

At a high level, the system architecture consists of:
SaaS Platform – Cloud-hosted platform providing procurement, invoicing, catalog management, and workflow automation.

Middleware / ERP Adapter – Java-based integration adapters deployed using Eclipse IDE, with plug-ins managed and component generators configured. It handles API calls, message transformation, routing, CDML definitions, and non-functional code for operational control.

ETL Pipeline – Performs extraction from CSV files, databases, and APIs via Confluent JDBC connectors; transforms data using filtering, lookup tables, and syntax rules; loads data into target tables. Validation is performed using Python scripts and libraries such as pandas, good tables, and engarde.

Service Deployment Layer – OSGi-based environment managing container services, container proxies, and arrays of component instances within a Java Virtual Machine. Services are deployed locally or centrally and integrated with the **OC4J server** and **Control Center Agent**.

Monitoring & Notification Module – Provides real-time status updates, event notifications, and data validation alerts using JSON messages secured with secret keys.

Figure 1: High-Level System Architecture
(Placeholder: SaaS ↔ Middleware/ERP ↔ ETL ↔ Database / Reporting Modules)

Key Components / Modules

Procurement Module – Manages requisitions, purchase orders, and supplier catalogs.

Invoicing Module – Handles invoice creation, approval, and ERP integration.

ETL Module – Manages extraction, transformation, and loading; includes staging area, data filtering, standardization, and aggregation.

API Module – Supports POST, PATCH, DELETE operations for transactional data, account imports, and exchange rate synchronization.

Deployment & Configuration Module – Uses OSGi and JVM to manage services, configuration files (prod_config, dev_config, qa.config), and local service instances.

Request & Workflow Management Module – Handles feature requests, template updates, and business object data via JSON and SQL (OPENJSON, OPENROWSET) for validation and tabular conversion.

Communication Flow / Interfaces

ERP and source system data pass through middleware adapters to APIs.
ETL pipelines extract, transform, and validate operational and transactional data before loading into target systems.
Notifications, alerts, and transactional messages are transmitted via secure JSON APIs using secret key authentication.
Service deployment and instance management are centralized via OSGi, enabling dynamic updates and scaling of components.

Supported Platforms & Environments

Operating Systems: Windows Server, Linux (Ubuntu, CentOS)
Databases: Oracle, SQL Server, MySQL
Middleware & Tools: Eclipse IDE, Java, Python (pandas, goodtables, engarde), OSGi framework
Cloud Environment: SaaS-hosted platform

Dependencies

- Internet connectivity for API communication and cloud access.
- Access credentials for ERP and database systems.
- Required software: Java runtime, Eclipse IDE, Python libraries, OSGi runtime.
- ETL connectors and drivers: JDBC, Confluent connectors, CSV parser libraries.

6. Prerequisites

Before beginning the integration of the SaaS P2P system, ensure the following requirements are met to guarantee a smooth setup, configuration, and deployment process.

6.1 Hardware Requirements

Client/Integration workstation: Minimum 8 GB RAM, 4-core processor, 100 GB free disk space.
Middleware / Integration server: Minimum 16 GB RAM, 8-core processor, 500 GB storage for staging, logs, and temporary ETL files.
Database server: Minimum 16 GB RAM, 8-core CPU, configured for high I/O performance.

6.2 Software / OS Requirements

Operating Systems: Windows Server 2016 or later, Linux Ubuntu 20.04 / CentOS 8.
Java Runtime Environment (JRE 8 or higher) for OSGi and middleware adapters.
Eclipse IDE with required plug-ins for integration development.
Python 3.x with libraries: pandas, goodtables, engarde for ETL validation.
ETL tools and connectors: Confluent JDBC connectors for database access.

6.3 Network Requirements / Ports

Stable internet connectivity for SaaS API communication.
Open ports for OSGi service deployment and middleware communication (default HTTP/HTTPS ports: 80/443, additional middleware ports as required).
Firewall rules allowing access to ERP endpoints, database servers, and cloud APIs.

6.4 Access / Permission Requirements

User credentials for SaaS platform.
ERP system user accounts with read/write permissions for required modules (accounts, invoices, catalogs).
Database access with privileges to read/write ETL staging and target tables.
Control Center Agent and OC4J server access for service deployment and monitoring.

6.5 Libraries & Connectors

JDBC drivers for Oracle, SQL Server, or MySQL.
API keys and secret keys for secure JSON communication between SaaS and middleware.
Middleware plug-ins, component generators, and configuration templates.

7. Installation & Setup

This section provides comprehensive instructions for installing and configuring the **SaaS P2P integration environment**, including middleware, ETL pipelines, service deployment, and validation. Following these steps ensures a consistent and reliable integration.

7.1 Step-by-Step Installation Instructions

Middleware & IDE Setup
Install **Eclipse IDE** on the integration workstation. Enable all required plug-ins for middleware adapters.
Install **Java Runtime Environment (JRE 8 or higher)**. Configure system environment variables (JAVA_HOME, PATH).
Import the middleware integration project into Eclipse. Verify that the **component generator** and **CDML definitions** are loaded.
Configure ERP connection parameters, including hostnames, ports, and authentication credentials.

Service Deployment
Install **OSGi runtime** on the integration/server machine.
Deploy container services, container proxies, and arrays of component instances.
Link OSGi services with the **OC4J server** and **Control Center Agent**.
Place configuration files (prod_config, dev_config, qa.config) in the designated directories.

ETL Setup
Install Python 3.x and required libraries: pandas, goodtables, engarde.
Configure ETL pipelines to extract data from source systems (CSV files, databases via JDBC, or APIs).
Define transformation rules using lookup tables, filters, and aggregation syntax.
Set staging and target directories for data loading.
Execute a test ETL run to verify data extraction, transformation, and loading.

Validation of Installation
Verify connectivity between middleware and APIs (POST, PATCH, DELETE).
Confirm OSGi services are running and can dynamically add new instances.
Ensure ETL pipelines execute without errors and load data correctly.
Confirm all configuration files and credentials are correctly referenced.

7.2 Directory / File Structure

Component	Directory / File Path	Description
Middleware Project	C:\Eclipse\Workspace_Middleware\	Eclipse project with integration adapters and plug-ins
OSGi Services	C:\OSGi\\Services\	Container services and proxies
Configuration Files	C:\OSGi\\Config\prod_config	Prod/dev/QA configurations
ETL Staging	D:\ETL\Staging\	Temporary data extracted from source systems
ETL Target	D:\ETL\Target\	Processed data ready for ERP or reporting
Logs	D:\ETL\Logs\	Error, execution, and service logs

7.3 Configuration Settings (Default vs Recommended)

Parameter	Default	Recommended	Notes
ERP Host	erp.example.com	Customer-specific host	Middleware endpoint
ERP Port	8080	443 (HTTPS)	Secure connection
API Timeout	30 sec	60 sec	For large datasets
ETL Batch Size	1000 rows	5000 rows	Performance tuning
OSGi JVM Memory	2 GB	8 GB	Supports multiple container instances

7.4 Environment Variables / Paths

JAVA_HOME → C:\Program Files\Java\jdk1.8.x
PATH → includes JAVA_HOME\bin and Eclipse executable path
PYTHONPATH → directory containing ETL scripts and Python libraries
OSGi runtime path → C:\OSGi\\Services
ETL staging & target paths → D:\ETL\Staging, D:\ETL\Target

7.5 Troubleshooting Common Installation Issues

Issue	Probable Cause	Resolution
Middleware cannot connect to ERP	Incorrect host, port, or credentials	Verify endpoint, check firewall, validate login
OSGi service fails to start	JVM memory insufficient or missing container	Increase JVM memory, verify service deployment
ETL extraction fails	JDBC driver missing or incorrect source path	Install correct JDBC driver, check CSV/API source path
Python ETL script errors	Missing libraries or incorrect paths	Install pandas, goodtables, engarde; verify PYTHONPATH
Configuration file not loaded	File path incorrect or syntax error	Confirm correct directory and valid syntax

8. Configuration

This section details the configuration required to enable successful integration between the **SaaS P2P platform** and ERP/middleware systems. Proper configuration ensures accurate API communication, data validation, and ETL execution.

8.1 Middleware Configuration

Configure middleware adapters in Eclipse with ERP endpoint URLs, ports, and authentication credentials.
Load required **component generators**, CDML definitions, and nonfunctional codes for service execution.
Specify container and proxy service instances in the OSGi runtime.

8.2 API Configuration

Configure API endpoints for transactional operations: POST, PATCH, DELETE.
Define authentication using secret keys for secure JSON communication.
Map API fields to middleware or ETL data parameters.
Set notification channels for real-time event messages.

8.3 ETL Configuration

Define staging and target tables in configuration files.
Configure extraction rules (source paths, database queries, CSV/API input).
Set transformation rules using lookup tables, filters, and aggregation parameters.
Define validation rules for data quality checks (pandas, goodtables, engarde).

8.4 Environment Variables

Verify JAVA_HOME, PATH, PYTHONPATH, and OSGi runtime paths.
Ensure ETL script paths and configuration files are correctly referenced.

8.5 Recommended Settings

Use HTTPS for all API calls.
Enable logging for middleware, ETL, and OSGi services.
Maintain separate configurations for **dev, QA, and prod environments**.

9. Integration Steps

This section provides detailed, step-by-step instructions for integrating the **SaaS P2P system** with ERP and backend systems, covering API calls, ETL mapping, validation, error handling, rollback, and logging.

9.1 Pre-Integration Validation

Before starting the integration, ensure the following:
Environment readiness: All prerequisites (hardware, software, network, access permissions) are verified.
Middleware & ETL check: Eclipse IDE, middleware adapters, OSGi services, ETL scripts, and configuration files are installed and tested.
Connectivity verification: Test API endpoints, ERP connections, database access, and OSGi service availability.
Backup: Backup existing configuration files, ETL scripts, and middleware settings.
Test datasets: Prepare a small subset of data for initial validation to avoid affecting production.

9.2 Step-by-Step Integration Procedure

Establish Middleware Connections
Configure the middleware adapter in Eclipse with ERP endpoint, authentication, and ports.
Verify connectivity using sample API calls (POST for invoices, GET for accounts).
Deploy OSGi Services
Start container services and proxies.
Register services with the **Control Center Agent**.
Confirm dynamic addition of new instances if required.
Configure ETL Pipelines
Extract data from ERP tables, CSV files, or APIs using Confluent JDBC connectors.
Transform data using lookup tables, filtering, and aggregation rules.
Load transformed data into target tables.
Map API and Data Fields
Define field mapping between ERP, middleware, ETL staging, and P2P payloads.
Map invoice lines, accounts, exchange rates, and business objects using JSON structures.
Execute Test Integration Cycle
Run ETL on a small test dataset.
Trigger API calls (POST, PATCH, DELETE) and verify success.
Confirm notifications and logging for errors.

9.3 API / SDK Calls / Commands

POST Invoice Example

```
POST /api/invoices
Host: saas..com
Authorization: Bearer <secret_key>
Content-Type: application/json

{
  "invoiceId": "INV-001",
  "supplierId": "SUP-123",
  "amount": 1500.00,
  "currency": "USD",
  "date": "2026-02-13"
}
```

Python ETL Snippet

```
import pandas as pd
from goodtables import validate

data = pd.read_csv('staging/invoices.csv')
validation = validate(data)
if validation['valid']:
    data.to_sql('target_invoices', con=engine, if_exists='append')
```

SQL OpenJSON Mapping

```
SELECT *
FROM OPENJSON(@jsonData)
WITH (
    invoiceId NVARCHAR(50),
    supplierId NVARCHAR(50),
    amount DECIMAL(10,2),
    currency NVARCHAR(10)
)
```

9.4 Data Flow & Mapping

Source → Middleware
ERP, CSV, and API data extracted into staging tables.
Middleware → SaaS
Data transformed, mapped, and sent as JSON payloads to APIs.
ETL Pipeline → Target
ETL validates, cleans, aggregates, and loads data into target tables for reporting or reconciliation.
Notifications & Logging
JSON notifications confirm successful transactions.

Logs capture errors, warnings, and processing metrics.

9.5 Testing Each Step for Verification

Connection Testing: Confirm API endpoints and database connections are reachable.
Data Validation: Compare ETL outputs with expected values.
Integration Testing: Execute a full procurement → invoice → account cycle.
Notification Verification: Confirm event notifications reach intended recipients.
Post-Integration Verification: Check final target tables, logs, and SaaS records for completeness and accuracy.

9.6 Error Handling & Logging

Error Type	Cause	Resolution
API Call Failure	Incorrect endpoint or authentication	Verify URL, secret key, and headers
ETL Validation Error	Missing fields, wrong data types	Correct source data, update mapping
OSGi Service Not Running	JVM memory insufficient, missing container	Increase JVM memory, restart service
Middleware Timeout	Network/firewall issue	Check ports, connectivity, VPN
Configuration Load Error	Incorrect syntax	Validate config files and paths

Logging Best Practices:
Enable debug mode for initial runs.
Maintain separate logs for middleware, ETL, and OSGi services.
Include timestamps, transaction IDs, and error codes.
Rotate logs regularly.

9.7 Rollback & Recovery Procedures

ETL Rollback
Restore target tables from backup before a failed ETL run.
Re-run ETL after correcting data or transformation issues.
Middleware Rollback
Revert middleware configuration files to last known good version.
Restart services and confirm API connectivity.
OSGi Service Recovery
Stop affected container services.
Redeploy using backup configuration and service packages.

Validate instance registration in Control Center Agent.
Data Correction & Retry
Correct invalid payloads or missing fields.
Re-trigger API calls with corrected data.
Confirm transaction success in SaaS and target systems.

9.8 Additional Recommendations

Maintain separate **dev, QA, and production environments**.
Schedule periodic ETL and API validation cycles.
Document all customizations in middleware, ETL, or OSGi.
Keep backups of all configuration files, scripts, and service packages.

10. Troubleshooting & Maintenance

This section provides detailed guidance on identifying, resolving, and preventing common issues during the integration and operation of the **SaaS P2P system**. It also outlines best practices for ongoing maintenance to ensure reliable and efficient performance.

10.1 Common Issues & Troubleshooting

Issue	Possible Cause	Resolution
Middleware connection failure	Incorrect endpoint, port, or credentials	Verify ERP/ API endpoints, port accessibility, and authentication keys
API call errors (POST/PATCH/DELETE)	Invalid payload structure or missing fields	Check JSON structure, validate required fields, re-run API call
ETL extraction/transform errors	Missing libraries, wrong file path, corrupted source data	Ensure Python libraries (pandas, goodtables, engarde) are installed; verify CSV/API sources
OSGi service startup failure	JVM memory insufficient, missing container	Increase JVM memory allocation, restart container services
Configuration file not loading	Syntax errors, wrong path	Validate config syntax, check file location, verify permissions
Data mismatch in target tables	Mapping errors or transformation issues	Review ETL transformation rules, check field mappings, re-run ETL pipeline
Notification failures	Misconfigured webhook or secret key	Verify JSON notification endpoint, ensure secret key matches

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10.2 Preventive Maintenance

Middleware & OSGi Services
Monitor CPU and memory usage on middleware and service servers.
Regularly restart OSGi services to clear stale container instances.
Apply software updates and patches for the middleware and runtime environment.
ETL Pipelines
Validate source data daily before extraction.
Monitor ETL execution logs for warnings or errors.
Schedule regular batch runs during off-peak hours to avoid system congestion.
Database Maintenance
Backup staging and target tables regularly.
Archive old data periodically to maintain optimal performance.
Validate indexes, primary keys, and constraints to prevent loading errors.

Configuration Files & Environment
Keep a versioned backup of all config files (prod_config, dev_config, qa.config).
Verify environment variables (JAVA_HOME, PYTHONPATH, OSGi paths) after updates or server changes.
Maintain consistent configurations across dev, QA, and production environments.

10.3 Monitoring & Logging Best Practices

Enable detailed logging for middleware, ETL, and OSGi services.
Include timestamps, transaction IDs, and error codes in logs for traceability.
Monitor logs daily for recurring warnings or errors.
Implement automated alerts for critical failures, such as API timeouts or ETL pipeline errors.
Archive logs weekly to prevent storage issues and maintain historical records.

10.4 Troubleshooting Procedures

Identify the Issue
Review logs, notifications, and system alerts to pinpoint the affected component (middleware, ETL, OSGi, API).
Confirm whether the issue is environment-specific or system-wide.
Isolate & Replicate
Use test datasets to reproduce the issue in dev/QA before applying fixes in production.
Confirm network connectivity, credentials, and service availability.
Apply Fix & Validate
Correct configuration errors, code issues, or data discrepancies.
Re-run ETL, API calls, or middleware services.
Validate results against expected outputs.
Escalation & Documentation
If unresolved, escalate to senior technical staff or vendor support.
Document the root cause, resolution steps, and preventive measures.

10.5 Best Practices for Long-Term Maintenance

Maintain separate **development, QA, and production environments**.
Schedule regular **ETL validation and API testing cycles**.
Keep middleware, OSGi, and ETL software updated with security patches.
Implement version control for scripts, configuration files, and transformation rules.
Regularly conduct **data quality audits** to ensure integrity across SaaS and ERP systems.
Train staff on troubleshooting procedures and maintain a **knowledge base** for recurring issues.

10.6 Optional Advanced Maintenance Tips

Automate ETL and API validation with cron jobs or scheduler tools.
Use monitoring dashboards to visualize API calls, ETL execution, and service health.
Apply load testing periodically to ensure the system can handle peak transactional volumes.
Maintain rollback scripts and backup snapshots for quick recovery in case of critical failures.

11. Troubleshooting & FAQ

This section consolidates common issues, provides practical guidance for resolving them, and offers tips for proactive monitoring. It is intended to help integrators, system admins, and developers quickly identify and resolve problems.

11.1 Common Errors & Solutions

Error	Cause	Solution
Middleware cannot connect to ERP	Incorrect endpoint, port, or authentication	Verify ERP endpoint URL, credentials, and firewall rules
API call fails (POST/PATCH/DELETE)	Invalid JSON payload or missing fields	Validate JSON structure, required fields, and secret keys
ETL extraction errors	Missing libraries, incorrect file path, corrupted data	Install required Python libraries (pandas, goodtables, engarde), verify source path and data integrity
OSGi service not starting	Insufficient JVM memory or missing container	Increase JVM memory, restart container services
Configuration file not loading	Wrong path or syntax errors	Validate file paths, file names, and syntax in prod_config, dev_config, or qa.config
Data mismatch in target tables	Mapping errors or transformation issues	Review ETL mapping rules and field mappings; re-run ETL pipelines

11.2 Log Analysis Tips

Enable **debug or verbose mode** for initial integration cycles to capture detailed logs.
Check **middleware, OSGi, and ETL logs** separately for pinpointing the root cause.
Look for **error codes, timestamps, transaction IDs**, and warning messages.
Maintain **archived logs** to track recurring issues over time.
Use **search and filter functions** in log viewers to quickly identify failed transactions or exceptions.

11.3 Contact / Support Information

Internal Support:
Integration Team Email: integration-support@company.com
Phone: +1-XXX-XXX-XXXX
Vendor / SaaS Support:
Support Portal: https://support..com
Email: help@.com
Phone (North America): +1-800--1

Escalation Guidelines:
Level 1: Integration Team → troubleshoot basic middleware, ETL, and API errors
Level 2: Vendor Support → unresolved system or SaaS platform issues
Level 3: Senior Technical Staff → critical failures affecting production

11.4 FAQs

Q: What should I do if an API call succeeds but data does not appear in SaaS?
A: Verify field mappings, ETL transformations, and ensure target tables are updated correctly.
Q: How often should ETL pipelines be validated?
A: Daily for production environments; weekly for dev/QA.
Q: Can I add custom fields to the API payload?
A: Yes, but ensure the middleware and ETL scripts are updated accordingly, and perform a test run before production deployment.
Q: What if I need to rollback a failed integration batch?
A: Restore target tables from backup, correct any errors, and re-run ETL and API calls.

11.5 Additional Tips

Keep **pre-defined templates** for configuration and ETL scripts to simplify future deployments.
Document **custom mapping rules** for future reference.
Maintain a **knowledge base** of recurring issues, fixes, and preventive measures.
Use **automation and monitoring tools** for proactive alerting on integration failures.

12. Maintenance & Updates

This section outlines procedures for maintaining the SaaS P2P integration environment, applying updates safely, and ensuring system continuity.

12.1 Upgrade Procedures

Follow a **staged upgrade approach**: dev → QA → production.
Backup all middleware, ETL scripts, OSGi configurations, and SaaS integration settings before upgrades.
Apply updates to middleware adapters, OSGi services, and ETL scripts sequentially.
Validate functionality after each upgrade stage using test datasets.

12.2 Backup & Restore Guidelines

Perform **daily backups** of ETL staging and target databases.
Maintain **versioned backups** of configuration files (prod_config, dev_config, qa.config) and middleware scripts.
Test restore procedures periodically to ensure data integrity and recovery readiness.

12.3 Monitoring & Alerts

Enable logging for all middleware, ETL, and OSGi services.
Configure **automated alerts** for failed API calls, ETL errors, or service downtime.
Periodically review logs and metrics to identify recurring issues proactively.

12.4 Version Compatibility Notes

Confirm **ERP and SaaS API versions** are compatible with the installed middleware and ETL scripts.
Maintain a compatibility matrix for all environments (dev, QA, production).
Before applying upgrades, verify dependent libraries, Python packages, and JVM versions.

12.5 Optional Advanced Maintenance

Automate ETL validation and API test runs using schedulers or cron jobs.
Maintain a **rollback plan** for critical updates.
Conduct periodic **load and performance testing** to ensure the system handles peak transaction volumes.

13. Appendices & References

This section provides supporting material to enhance understanding and implementation of the SaaS P2P integration, including sample files, diagrams, API references, and external documentation.

13.1 Glossary of Terms

Term	Definition
SaaS	Software as a Service – cloud-based software accessed via subscription
P2P	Purchase-to-Pay – end-to-end process for procurement and invoice management
ETL	Extract, Transform, Load – process for moving data from source to target systems
OSGi	Open Services Gateway initiative – a framework for modular Java-based services
Middleware	Software that connects ERP systems to SaaS applications
JSON	JavaScript Object Notation – lightweight data exchange format
CDML	Component Definition Markup Language – used for defining reusable service components

13.2 Acronyms

Acronym	Meaning
ERP	Enterprise Resource Planning
API	Application Programming Interface
CSV	Comma-Separated Values
JVM	Java Virtual Machine
QA	Quality Assurance
PROD	Production Environment
DEV	Development Environment

13.3 Sample Configuration Files

Example: prod_config.json

```
{
  "middleware_endpoint": "https://middleware.company.com/api",
  "erp_endpoint": "https://erp.company.com/api",
  "secret_key": "********",
  "staging_db": "staging_db_prod",
  "target_db": "target_db_prod",
  "log_level": "DEBUG",
  "osgi_container": "container1"
}
```

Example: ETL Mapping File (mapping.csv)

Source Field	Target Field	Transformation
invoice_id	invoiceId	None
supplier_id	supplierId	Uppercase
amount	totalAmount	Round(2)
currency	currencyCode	None

13.4 Diagrams

System Architecture Diagram – shows SaaS, middleware, ETL, and ERP connections.
Data Flow Diagram – illustrates extraction, transformation, API integration, and loading into target tables.
Deployment Diagram – shows OSGi container services, control center agents, and server environments.

13.5 API Reference Tables

API	Method	Endpoint	Payload	Response
Create Invoice	POST	/api/invoices	JSON	201 Created / Error
Update Invoice	PATCH	/api/invoices/{id}	JSON	200 OK / Error
Delete Invoice	DELETE	/api/invoices/{id}	None	204 No Content / Error
Get Accounts	GET	/api/accounts	Query params	JSON / Error

13.6 Sample Scripts

Python ETL script for loading invoices:

```
import pandas as pd
from sqlalchemy import create_engine

engine = create_engine('postgresql://user:pass@host:5432/db')
data = pd.read_csv('staging/invoices.csv')
data.to_sql('target_invoices', con=engine, if_exists='append', index=False)
```

Middleware health check script

```
curl -X GET https://middleware.company.com/healthcheck
```

13.7 Reference Links / Documentation

SaaS API Documentation: https://api..com/docs
ISO 26511: Information management in software lifecycle
ISO 27001: Information Security Management System compliance
Middleware vendor manuals and OSGi framework guides

Optional Additional Appendices

Change Log: Tracks configuration updates, API version changes, and ETL pipeline modifications.
FAQ Reference: Links to frequently asked integration questions.
Environment Matrix: Details dev, QA, and production environment configurations.